Chemistry Mock

Answers and Marking Scheme

M: method mark A: answer mark (only if the related M mark is awarded) B: independent mark C: communication mark
Underlined words are key words that must appear in the answer. Chemical equations must be balanced; contradictory answers score no marks for that part. For revision only; other valid answers should also be credited.

Section A Answers

1	2	3	4	5	6	7	8	9	10	11	12
D	A	C	C	A	D	D	B	B	A	D	C
13	**14**	**15**	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**
A	A	B	C	A	C	B	A	B	B	B	A
25	**26**	**27**	**28**	**29**	**30**	**31**	**32**	**33**	**34**	**35**	**36**
D	B	B	C	D	C	D	D	D	A	C	C

1. D Atomic numbers 15 (P) and 33 (As) are both in Group V. 2 (He) is a noble gas while 12 (Mg) is in Group II; 7 (N) is in Group V while 17 (Cl) is in Group VII; 13 (Al) is in Group III while 21 (Sc) is a transition metal.

2. A X is connected to the positive terminal and Y to the negative terminal. Green $\mathbf{Ni^{2+}}$ ions migrate to the negative electrode Y and yellow $\mathbf{CrO_4^{2-}}$ ions migrate to the positive electrode X. Sucrose is a non-electrolyte: the gel has no mobile ions to carry the current, so the patches remain essentially unmoved.

3. C The reaction between magnesium hydroxide and stomach acid is neutralisation, with no change in oxidation numbers. $\mathbf{OCl^-}$ in bleach is an oxidising agent; vitamin C is a reducing agent (antioxidant); silver is oxidised to $\mathbf{Ag_2S}$.

4. C Acid in $\mathbf{25.00\ cm^3} = \frac{1.60}{192.0} \times \frac{25.00}{250.0} = \mathbf{8.33 \times 10^{-4}}$ mol; NaOH $= \mathbf{0.120 \times 0.02083 = 2.50 \times 10^{-3}}$ mol. n $= \frac{2.50\times10^{-3}}{8.33\times10^{-4}} = \mathbf{3}$.

5. A Moles of $\mathbf{Zn^{2+}} = \mathbf{0.30 \times 0.0200 = 0.0060}$; moles of $\mathbf{OH^-} = \mathbf{0.20 \times 0.0300 = 0.0060}$. As $\mathbf{Zn^{2+} : OH^- = 1 : 2}$, $\mathbf{OH^-}$ is limiting and consumes only 0.0030 mol of $\mathbf{Zn^{2+}}$. 0.0030 mol remains; concentration $= \frac{0.0030}{0.0500} = \mathbf{0.060\ M}$.

6. D Moles of $\mathbf{AgNO_3} = \frac{5.10}{170.0} = \mathbf{0.0300}$. If the residue is Ag: $\mathbf{0.0300 \times 108.0 = 3.24}$ g, which fits. AgO would give 3.72 g, $\mathbf{Ag_2O}$ 3.48 g and $\mathbf{AgNO_2}$ 4.62 g.

7. D The delivery tube should be taken out of the water BEFORE removing the burners; otherwise, as the set-up cools, the pressure falls and water sucks back into the hot combustion tube, which may crack. The products contain alkenes, which decolourise acidified potassium permanganate solution.

8. B Moles of Pb $= \frac{6.21}{207.0} = 0.0300$; mass of O $= 6.85 - 6.21 = 0.64$ g, moles of O $= 0.0400$. Pb : O $= 3 : 4$.

9. B $3C(s) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow CH_3COCH_3(l)$.
$\Delta H_f = 3(-393.5) + \frac{3}{2}(-571.6) - \frac{1}{2}(-3\,580) = -247.9\ \text{kJ mol}^{-1}$.

10. A $-CF_2-$ and $-CHCl-$ units alternate along the chain, so the repeating unit is $-CF_2-CHCl-$ and the monomer is $CF_2{=}CHCl$.

11. D X is connected to the positive terminal and is the anode. Silver is an active electrode and loses electrons more readily than OH^-, so it dissolves. Y is the cathode: H^+ (from water) gains electrons more readily than K^+, so hydrogen is evolved.

12. C X: $H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$; Y: $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$. OH^- is consumed at X and formed at Y, so it migrates towards X. The overall reaction produces water, so the solution becomes more dilute.

13. A CH_2Cl_2 is tetrahedral and its bond dipoles do not cancel out, so it is polar. BCl_3 is trigonal planar and symmetrical, hence non-polar; PH_3 has a lone pair and is pyramidal; SCl_2 has two lone pairs and is V-shaped.

14. A (1): Mn is +6 in K_2MnO_4 and +4 in MnO_2. (2): Mn is reduced and O ($-2 \rightarrow 0$) is oxidised, both from $KMnO_4$. (3): the two Mn atoms gain $1 + 3 = 4$ electrons, so 4 mol of electrons are transferred per mol of O_2; the statement is incorrect.

15. B (1) Magnesium is more reactive than iron and provides sacrificial protection. (2) Nickel is less reactive than iron; on contact, iron loses electrons more readily, so rusting is speeded up. (3) The chromium oxide layer on stainless steel protects the steel.

16. C (1) Both give white precipitates ($PbSO_4$, $PbCl_2$), so they cannot be distinguished. (2) Only sulphuric acid gives white $BaSO_4$. (3) With sulphuric acid, sparingly soluble $CaSO_4$ coats the limestone and effervescence soon stops; with hydrochloric acid, effervescence continues.

17. A The order of reactivity is S > R > T. (1) The oxide of S is not reduced by carbon, so S is more reactive. (2) R is more reactive than T and can displace it. (3) An oxide that decomposes on heating alone shows that T is the least reactive, so the statement is incorrect.

18. C X (cyclopentane) and Y (pent-2-ene) both have the molecular formula C_5H_{10}, so they are isomers. X is a cycloalkane and Y an alkene: they differ in functional group and are not in the same homologous series despite having the same general formula. Each carbon of the C=C bond in Y carries two different groups, so Y has cis-trans isomers.

19. B (1) No oxidation number changes; it is not a redox reaction. (2) O in H_2O_2 changes from -1 to -2; it is reduced, so H_2O_2 is an oxidising agent. (3) O in H_2O_2 changes from -1 to 0; it is oxidised, so H_2O_2 is a reducing agent.

20. A (1) Calcium hydroxide is an alkali and neutralises the acid in waste water. (2) Heating with NH_4Cl gives ammonia, whereas $NaCl$ does not react. (3) Calcium hydroxide solution contains water and cannot be a drying agent.

21. B (1) After rinsing with distilled water, the pipette must also be rinsed with the sample solution, or the solution will be diluted. (2) This is the correct way to read the pipette. (3) A pipette is calibrated to allow for the liquid left in the tip, so it should not be blown out.

22. B Condensation of steam releases heat. Thermal decomposition of limestone and dissolving ammonium nitrate in water are both endothermic.

23. B (1) $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$. (2) The –OH in ethanol is a covalently bonded hydroxyl group and does not ionise to give OH^-, so the solution is neutral. (3) Ethanol is oxidised and $Cr_2O_7^{2-}$ is reduced to green Cr^{3+}.

24. A Metals still have metallic bonding and delocalised electrons in the liquid state. These mobile electrons carry the current, so the 2nd statement explains the 1st.

25. D Halving the volume immediately doubles all concentrations, so $[NH_3]$ rises to $2c$. The equilibrium position then shifts to the right (fewer gas molecules), so $[NH_3]$ continues to rise and finally steadies at a level above $2c$.

26. B For the reverse reaction, $K_c = \frac{1}{64.0}$; halving the coefficients takes the square root: $K_c = \sqrt{\frac{1}{64.0}} = 0.125$.

27. B At equilibrium, 0.40 mol of PCl_5 remains. $[PCl_5] = 0.20\ M$, $[PCl_3] = [Cl_2] = 0.10\ M$.
$K_c = \frac{0.10 \times 0.10}{0.20} = 0.050\ mol\ dm^{-3}$.

28. C Volume of $CO_2 = 240\ cm^3$, i.e. 0.0100 mol. NaOH required $= 0.0200$ mol; volume $= \frac{0.0200}{0.20} = 0.100\ dm^3 = 100\ cm^3$.

29. D Excess HBr adds across both C=C bonds. By Markovnikov's rule, H adds to the carbon already bearing more hydrogen atoms and Br to the other carbon, giving IV.

30. C P has a C=C bond and can undergo addition polymerisation. Q has both $-NH_2$ and $-COOH$ groups and can condense on its own to give a polyamide. R has only two $-NH_2$ groups and needs another monomer (e.g. a dicarboxylic acid) for condensation polymerisation.

31. D IV has a five-carbon main chain with the carbonyl group at C-3 and the methyl group at C-2. I should be ethyl propanoate; in II the chlorine is on C-2, so it is 2-chloro-2-methylpropane; III should be numbered to give the hydroxyl group the lowest locant: but-3-en-2-ol.

32. D Step 1 reduces a carboxylic acid to a primary alcohol, which requires $LiAlH_4$; $NaBH_4$ cannot reduce carboxylic acids. Step 2 dehydrates the alcohol to an alkene, which requires conc. sulphuric acid and heating.

33. D A shows variable oxidation states (+2, +7); B shows catalytic property; C shows the formation of coloured ions. Reacting with dilute acid is a property of reactive metals in general, not a characteristic of transition metals.

34. A (1) The concentrations are the same and Experiment II is at a higher temperature, so its initial rate is higher. (2) Experiment II has only 0.020 mol of NaOCl, half of that in Experiment I, so the volume of oxygen is halved. (3) The mass of a catalyst is unchanged after the reaction.

35. C (1) No gas is formed, so the mass does not change. (2) H^+ is consumed, so the pH rises. (3) I_2 is formed, so the colour intensity increases.

36. C The carbon bearing the hydroxyl group is bonded to H, OH and two identical $CH(CH_3)_2$ groups, and no other carbon has four different groups, so there is no chiral centre and the 1st statement is false. The carbon bearing the hydroxyl group is bonded to two carbon atoms, so it is a secondary alcohol and the 2nd statement is true.

Section B Marking Scheme

1. (7 marks)

(a) (1)

B1 I and Cl share one pair of electrons, each atom having three lone pairs (8 outermost electrons).

(b) (1)

B1 A chlorine atom is smaller, so the attraction between its nucleus and an incoming electron is stronger.

(c)(i) (1)

B1 The solution changes from pale yellow-green to colourless.

(c)(ii) (1)

B1 $SO_2 + Cl_2 + 2H_2O \rightarrow 4H^+ + 2Cl^- + SO_4^{2-}$

(c)(iii) (3)

B1 The white precipitate in X is barium sulphate, showing sulphate ions were formed; W shows SO_2 alone gives no sulphate ions.

B1 The oxidation number of sulphur increases from +4 in SO_2 to +6 in SO_4^{2-}.

B1 SO_2 is oxidised (it reduces chlorine to chloride ions), so it is a reducing agent.

2. (7 marks)

(a) (1)

B1 All the M ions have been precipitated (Na_3PO_4 is in excess).

(b)(i) (1)

B1 To remove soluble ions (e.g. Na^+, Cl^-) on the precipitate; otherwise the mass obtained would be too high.

(b)(ii) (2)

B1 The line follows the same initial slope and levels off at 8 cm^3.

B1 The final mass is 0.62 g.

(c) (3)

M1 No. of moles of M ions = $0.30 \times 0.0400 = 0.0120$

M1 No. of moles of PO_4^{3-} at the end point = $0.50 \times 0.0160 = 0.00800$

A1 M : PO_4 = 3 : 2, so the empirical formula is $M_3(PO_4)_2$.

3. (8 marks)

(a) (2)

B1 $AlCl_3$: trigonal planar.

B1 PCl_3: pyramidal, with a lone pair on P.

(b)(i) (2)

B1 Al in $AlCl_3$ has only six outermost electrons; a Cl atom of another $AlCl_3$ donates a lone pair to it.

B1 Structure: two Al atoms linked by two bridging Cl atoms, with one arrow from each bridging Cl to Al.

(b)(ii) (2)

B1 Both have simple molecular structures; boiling involves overcoming intermolecular forces.

B1 PCl_3 has a larger molecular size, so its van der Waals' forces are stronger.

(b)(iii) (2)

B1 Atoms within each layer are held by strong covalent bonds.

B1 Layers are held by weak van der Waals' forces, so they can slide over one another.

4. (7 marks)

(a) (1)

B1 Calcium oxide / calcium hydroxide is only sparingly soluble, so the reaction is slow and the end point is not sharp.

(b) (1)

B1 Methyl orange (or phenolphthalein).

(c) (1)

B1 Pipette (accept burette).

(d) (1)

B1 No change: the number of moles of excess acid is unchanged by adding water.

(e) (3)

M1 No. of moles of excess HCl = 0.100 × 0.01860 = 0.00186

M1 HCl reacted = 0.0150 − 0.00186 = 0.01314 mol; CaO = 0.00657 mol

A1 Percentage = 0.00657 × 56.1 ÷ 0.500 × 100% = 73.7%

5. (7 marks)

(a) (1)

B1 HOOC–CH2–COOH

(b)(i) (2)

B1 Only the two hydrogen atoms in the carboxyl groups are ionisable.

B1 The two hydrogen atoms bonded to carbon do not ionise in water.

(b)(ii) (2)

M1 Heat released = 75.0 × 4.20 × 8.30 = 2 615 J; moles of water = 0.0500

A1 $\Delta H = -2.615 \div 0.0500 = -52.3$ kJ mol^{-1}

(b)(iii) (2)

B1 Propanedioic acid is a weak acid and is only partially ionised.

B1 Some of the energy released is absorbed to ionise the acid further during neutralisation.

6. (7 marks)

(a) (2)

B1 Brine provides mobile ions to conduct electricity in the cell.

B1 Ions move through the wet paper, completing the circuit.

(b) (2)

B1 $Fe(s) \rightarrow Fe2+(aq) + 2e-$

B1 $O2(g) + 2H2O(l) + 4e- \rightarrow 4OH-(aq)$

(c) (1)

B1 The dotted line starts at a higher voltage than the solid line, because zinc is more reactive than iron.

(d)(i) (1)

B1 Rust (hydrated iron(III) oxide).

(d)(ii) (1)

B1 The rust covers the iron, reducing the area in contact with the electrolyte (or the iron/water is being used up).

7. (7 marks)

(a) (2)

B1 Mix the two solids in a beaker / polystyrene cup and stir.

B1 Use a thermometer to measure the temperature before and after mixing; the temperature falls.

(b) (1)

B1 Carry it out in a fume cupboard because ammonia is irritating / toxic.

(c)(i) (1)

B1 $N2(g) + 2H2(g) + 3/2\ O2(g) \rightarrow NH4NO3(s)\ \Delta H = -366$ kJ mol^{-1}

(c)(ii) (2)

M1 $\Delta H = [2(-468) + 2(-46.1) + (-393.5) + 11(-285.8)] - [(-4\ 081) + 2(-366)]$

A1 $\Delta H = +247.5$ kJ mol^{-1}

(c)(iii) (1)

B1 Heat is gained from the surroundings (or the reaction is incomplete, or some NH3 dissolves in the water formed, which is exothermic).

8. (6 marks)

(a) (6)

B1 Crude oil is separated by fractional distillation, based on the different boiling points of its components.

B1 A heavy fraction (e.g. heavy oil / kerosene) is cracked at high temperature with a catalyst such as aluminium oxide.

B1 Equation for cracking producing propene, e.g. C10H22 → C7H16 + C3H6.

B1 Propene is separated from the other cracking products (e.g. by fractional distillation).

B1 Propene undergoes addition polymerisation (catalyst, with heating / pressure); equation: n CH2=CHCH3 → –[CH2–CH(CH3)]n–

C1 Communication mark: complete sentences/paragraphs, logical sequence of steps, correct use of terms and equations.

9. (5 marks)

(a) (3)

M1 After mixing (before reaction): [Ag+] = 0.12 M and [Fe2+] = 0.080 M

M1 At equilibrium: [Ag+] = 0.10 M and [Fe2+] = 0.060 M

A1 Kc = [Fe3+] ÷ ([Ag+][Fe2+]) = 0.020 ÷ (0.10 × 0.060) = 3.33 $mol^{-1}\ dm^3$ (Ag(s) does not appear in the expression)

(b) (2)

B1 Kc decreases as the temperature rises, showing the equilibrium position shifts to the left on heating.

B1 Heating favours the endothermic direction, so the backward reaction is endothermic and the forward reaction is exothermic.

10. (6 marks)

(a)(i) (1)

B1 Substitution.

(a)(ii) (1)

B1 CH3CHBrCH3 + NaOH → CH3CH(OH)CH3 + NaBr

(b)(i) (1)

B1 CH3COOCH(CH3)2 + NaOH → CH3COONa + CH3CH(OH)CH3

(b)(ii) (2)

B1 Flask containing the mixture (with anti-bumping granules), with heating shown.

B1 Vertical condenser with water entering at the bottom and leaving at the top; the system must not be sealed (a sealed set-up scores 0).

(c) (1)

B1 –CH2–CH(CN)–

11. (7 marks)

(a) (1)

B1 Secondary.

(b) (1)

B1 2

(c)(i) (1)

B1 With J, the solution changes from purple to colourless; with L, there is no observable change.

(c)(ii) (1)

B1 L gives colourless gas bubbles, and the gas gives a pop sound with a burning splint; E shows no observable change.

(d) (3)

B1 Step 1: dilute H_2SO_4, heat under reflux (accept NaOH(aq), heat under reflux, then acidify).

B1 Intermediate: $CH_3CH(OH)CH_2COOH$.

B1 Step 2: concentrated H_2SO_4, heat (or Al_2O_3, heat). (Accept dehydration before hydrolysis, with the intermediate $CH_3CH{=}CHCOOCH_3$.)

12. (4 marks)

(a) (4)

B1 Iron shows variable oxidation states (+2 and +3), so it acts as a catalyst.

B1 Fe^{2+} is oxidised by $S_2O_8^{2-}$: $2Fe^{2+} + S_2O_8^{2-} \rightarrow 2Fe^{3+} + 2SO_4^{2-}$

B1 Fe^{3+} then oxidises I^- and is regenerated: $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$; this alternative pathway has a lower activation energy.

B1 Zn^{2+} cannot, because zinc has only one oxidation state in its compounds (+2).

13. (6 marks)

(a) (6)

B1 Use the same (excess) mass of magnesium ribbon, the same volume of acid and the same temperature; vary only the acid concentration.

B1 Tilt the flask so that the magnesium falls into the acid and start the stopwatch at once; record the volume of hydrogen in the gas syringe at regular time intervals until it stops changing.

B1 Sketch: the 0.20 M curve has the steeper initial slope.

B1 Acid is limiting: the final volumes are 96 cm^3 (0.20 M) and 48 cm^3 (0.10 M), i.e. they differ by a factor of two.

B1 Higher concentration gives more frequent effective collisions between H^+ ions and magnesium, hence a higher rate.

C1 Communication mark: complete sentences/paragraphs, logical sequence of steps, correct use of terms.